

## High Performance Computing Lab - MPI Assignments

Name : M Varshith Reddy

HT No : 2303A52087

Batch No : 40

### Assignment 1: Basic Point-to-Point Communication using MPI

**Scenario:** A distributed logging system needs to transfer status messages from a master process to a worker process.

#### Python Code (assignment1.py)

```
from mpi4py import MPI

def main():
    # Initialize the communicator
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()
    size = comm.Get_size()

    if size < 2:
        if rank == 0:
            print("Error: This program requires at least 2 processes.")
            return

    if rank == 0:
        # Master process
        message = "System status: ALL SYSTEMS GO!"
        # Send message to rank 1
        comm.send(message, dest=1, tag=11)
        print(f"[Rank 0] Sent message: '{message}' to Rank 1")

    elif rank == 1:
        # Worker process
        # Receive message from rank 0
        received_msg = comm.recv(source=0, tag=11)
        print(f"[Rank 1] Received message: '{received_msg}' from Rank 0")

if __name__ == "__main__":
    main()
```

## Execution Command

```
mpiexec -n 2 python assignment1.py
```

## Output

[Rank 0] Sent message: 'System status: ALL SYSTEMS GO!' to Rank 1

[Rank 1] Received message: 'System status: ALL SYSTEMS GO!' from Rank 0

## Assignment 2: Integer Data Exchange between Processes

**Scenario:** A parallel sensor network distributes integer readings from a control node to compute nodes.

### Python Code (assignment2.py)

```
from mpi4py import MPI
```

```
def main():
```

```
    comm = MPI.COMM_WORLD
```

```
    rank = comm.Get_rank()
```

```
    size = comm.Get_size()
```

```
    if size < 4:
```

```
        if rank == 0:
```

```
            print("Error: Please run with at least 4 processes.")
```

```
        return
```

```
    if rank == 0:
```

```
        # Send different integers to ranks 1, 2, and 3
```

```
        sensor_data = {1: 1024, 2: 2048, 3: 4096}
```

```
        for dest_rank, val in sensor_data.items():
```

```
            # Using the destination rank as the tag for differentiation
```

```
            comm.send(val, dest=dest_rank, tag=dest_rank)
```

```
            print(f"[Rank 0] Sent value {val} to Rank {dest_rank} with tag {dest_rank}")
```

```

elif rank in [1, 2, 3]:
    # Receive data matching this process's rank as the tag
    received_val = comm.recv(source=0, tag=rank)
    print(f"[Rank {rank}] Received sensor value {received_val} from Rank 0")

if __name__ == "__main__":
    main()

```

## Execution Command

```
mpiexec -n 4 python assignment2.py
```

## Output

```

[Rank 0] Sent value 1024 to Rank 1 with tag 1
[Rank 0] Sent value 2048 to Rank 2 with tag 2
[Rank 0] Sent value 4096 to Rank 3 with tag 3
[Rank 1] Received sensor value 1024 from Rank 0
[Rank 2] Received sensor value 2048 from Rank 0
[Rank 3] Received sensor value 4096 from Rank 0

```

## Assignment 3: One-to-One Communication Pattern

**Scenario:** Each process exchanges boundary data with its neighboring process (Even to Odd).

### Python Code (assignment3.py)

```

from mpi4py import MPI

def main():
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()
    size = comm.Get_size()

    if size % 2 != 0:
        if rank == 0:
            print("Error: Please run with an even number of processes.")
            return

    if rank % 2 == 0:

```

```

# Even ranks send data to the next odd rank (rank + 1)
dest = rank + 1
boundary_data = f"Boundary Matrix from Node {rank}"
comm.send(boundary_data, dest=dest, tag=99)
print(f"[Rank {rank} - EVEN] Sent data to Rank {dest}")

else:
    # Odd ranks receive data from the previous even rank (rank - 1)
    source = rank - 1
    received_data = comm.recv(source=source, tag=99)
    print(f"[Rank {rank} - ODD] Received: '{received_data}' from Rank {source}")

if __name__ == "__main__":
    main()

```

## Execution Command

```
mpiexec -n 4 python assignment3.py
```

## Output

```

[Rank 0 - EVEN] Sent data to Rank 1
[Rank 2 - EVEN] Sent data to Rank 3
[Rank 1 - ODD] Received: 'Boundary Matrix from Node 0' from Rank 0
[Rank 3 - ODD] Received: 'Boundary Matrix from Node 2' from Rank 2

```

# Assignment 4: Deadlock Analysis in MPI Send/Receive

**Scenario:** A parallel application experiences a hang due to improper ordering of communication calls.

## Deadlock Explanation

If two processes (Rank 0 and Rank 1) both execute `comm.send()` sequentially before executing `comm.recv()`, and if the messages are large enough to bypass standard system buffers, both processes will block indefinitely. They are both waiting for the other to start receiving, creating a **deadlock**.

To fix this, we reorder the calls: one process must initiate the send, while the other initiates the receive.

## Python Code (assignment4.py - Corrected)

```
from mpi4py import MPI

def main():
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()

    if comm.Get_size() < 2:
        return

    # To avoid deadlock, we ensure complementary communication ordering.
    # If Rank 0 sends then receives, Rank 1 MUST receive then send.

    if rank == 0:
        # 1. Rank 0 Sends first
        comm.send("Ping from 0", dest=1, tag=1)
        print("[Rank 0] Sent message to Rank 1")

        # 2. Rank 0 Receives second
        msg = comm.recv(source=1, tag=2)
        print(f"[Rank 0] Received: {msg}")

    elif rank == 1:
        # 1. Rank 1 Receives first
        msg = comm.recv(source=0, tag=1)
        print(f"[Rank 1] Received: {msg}")

        # 2. Rank 1 Sends second
        comm.send("Pong from 1", dest=0, tag=2)
        print("[Rank 1] Sent message to Rank 0")

if __name__ == "__main__":
    main()
```

## Execution Command

```
mpiexec -n 2 python assignment4.py
```

## Output

[Rank 0] Sent message to Rank 1  
[Rank 1] Received: Ping from 0  
[Rank 1] Sent message to Rank 0  
[Rank 0] Received: Pong from 1

## Assignment 5: Master-Worker Communication

**Scenario:** Master distributes tasks to workers and collects results.

### Python Code (assignment5.py)

```
from mpi4py import MPI
import time

def main():
    comm = MPI.COMM_WORLD
    rank = comm.Get_rank()
    size = comm.Get_size()

    if size < 2:
        if rank == 0:
            print("Error: Master-Worker model requires at least 2 processes.")
            return

    if rank == 0:
        # MASTER PROCESS
        print(f"[Master] Distributing tasks to {size - 1} workers...")

        # 1. Send tasks to all workers
        for worker_id in range(1, size):
            task_payload = worker_id * 5 # Example data
            comm.send(task_payload, dest=worker_id, tag=100)
            print(f"[Master] Sent task data ({task_payload}) to Worker {worker_id}")

        # 2. Collect results from all workers
        print("[Master] Waiting for results...")
        for worker_id in range(1, size):
            result = comm.recv(source=worker_id, tag=200)
            print(f"[Master] Collected result ({result}) from Worker {worker_id}")

    else:
```

```

# WORKER PROCESS
# 1. Receive task from master
task_data = comm.recv(source=0, tag=100)
print(f"[Worker {rank}] Received task data ({task_data}). Computing...")

# Simulate computation (e.g., squaring the number)
time.sleep(0.5)
computation_result = task_data ** 2

# 2. Send result back to master
comm.send(computation_result, dest=0, tag=200)
print(f"[Worker {rank}] Sent result ({computation_result}) back to Master")

if __name__ == "__main__":
    main()

```

## Execution Command

```
mpiexec -n 4 python assignment5.py
```

## Output

```

[Master] Distributing tasks to 3 workers...
[Master] Sent task data (5) to Worker 1
[Master] Sent task data (10) to Worker 2
[Master] Sent task data (15) to Worker 3
[Master] Waiting for results...
[Worker 1] Received task data (5). Computing...
[Worker 2] Received task data (10). Computing...
[Worker 3] Received task data (15). Computing...
[Worker 1] Sent result (25) back to Master
[Worker 2] Sent result (100) back to Master
[Worker 3] Sent result (225) back to Master
[Master] Collected result (25) from Worker 1
[Master] Collected result (100) from Worker 2
[Master] Collected result (225) from Worker 3

```